

Customer Journey

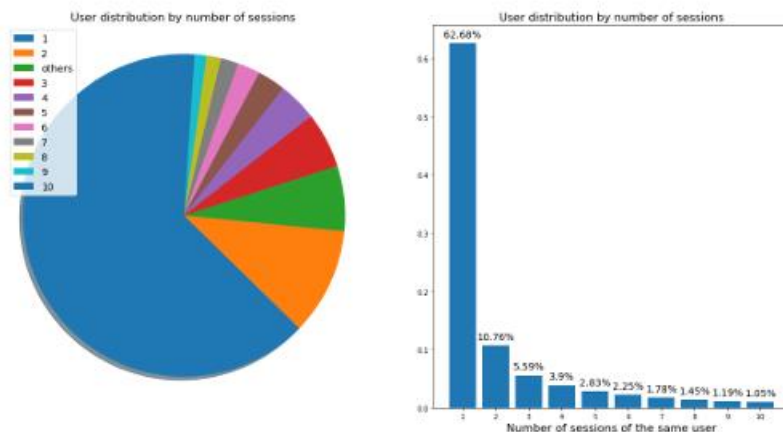
1. Background

This document analyses the user sessions during the month of September 2020. The analysis aims to identify which sessions are more important to understand how familiar users use applications.

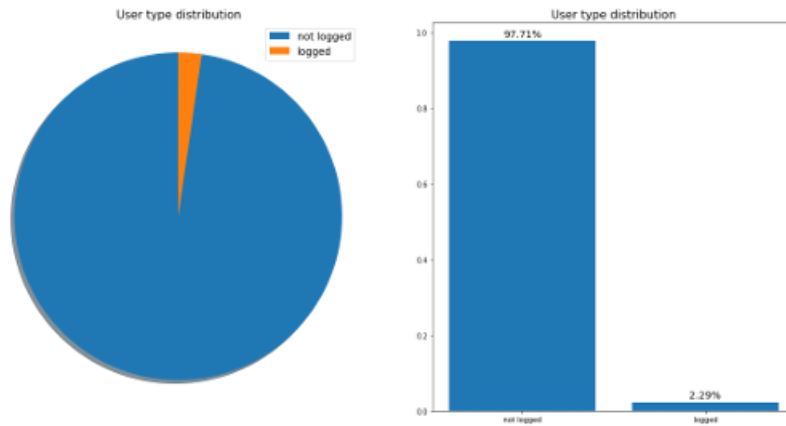
To achieve the most representative data possible, different filters have been applied to eliminate sessions that show non-human behaviours.

2. Initial Data Analysis

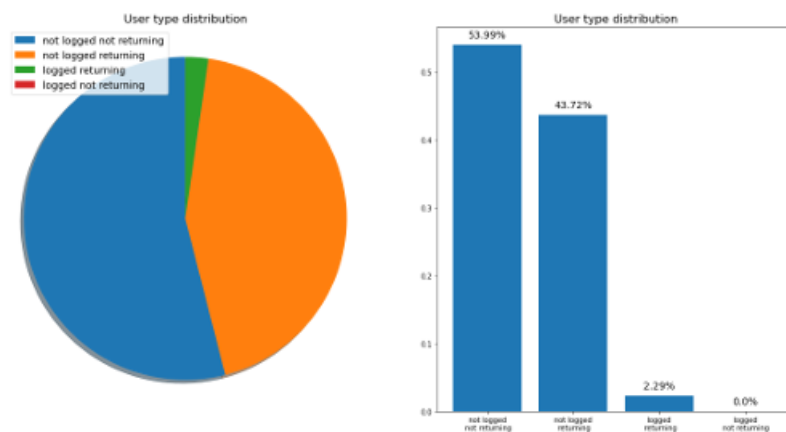
In this study, we work with user sessions from September 2020. The dataset contains 488.032 sessions from 355.902 different users and only 17% of users have registered more than four visits in the month of September.



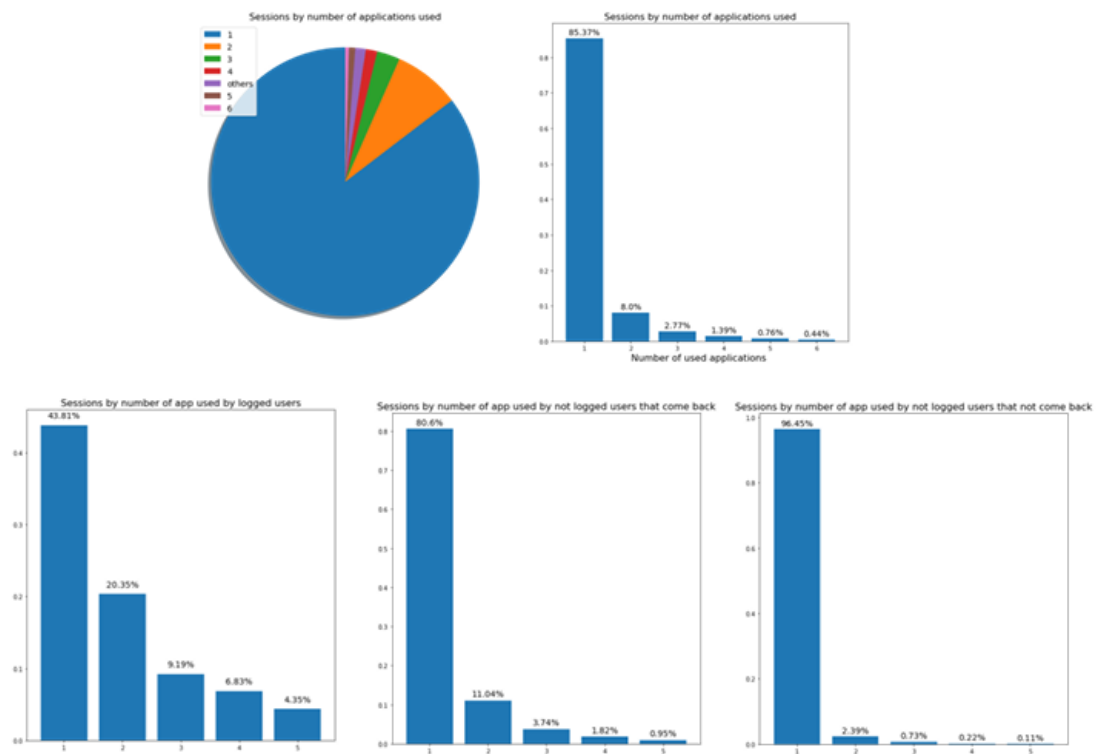
As we can see below, most of the users are not logged users, almost reaching the 98% of the users.



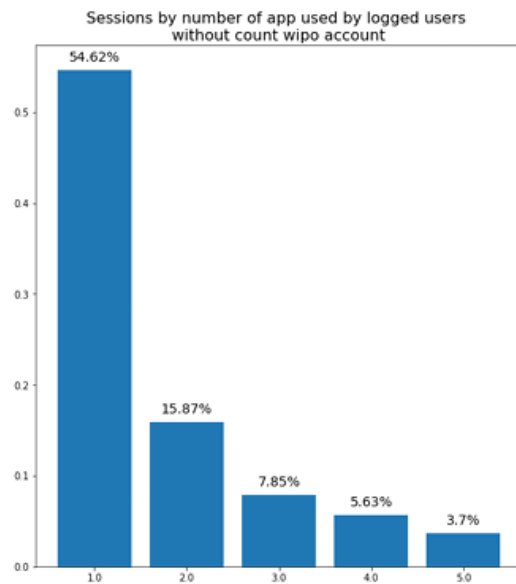
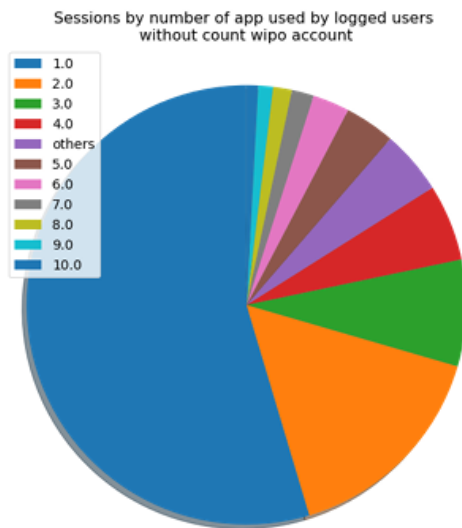
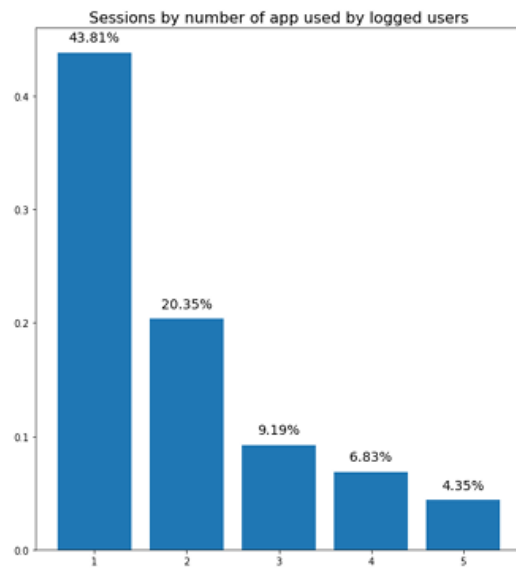
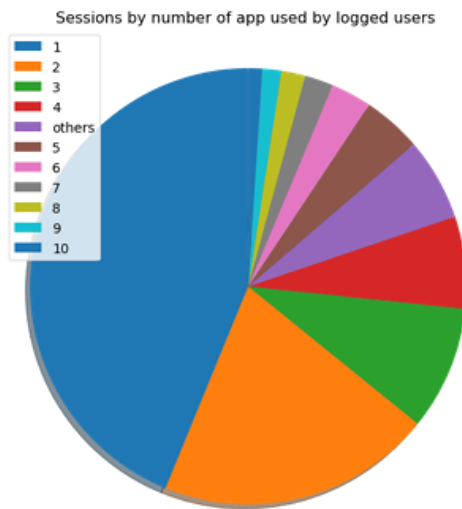
It could be expected that the non-logged in users do not have enough knowledge to use the applications correctly, but it has been observed that, as shown below, there is a large number of non-logged in users who visit the portal on multiple occasions. These users can be interesting because they can show an expert behaviour but different from that of registered users since the actions they can perform is reduced.



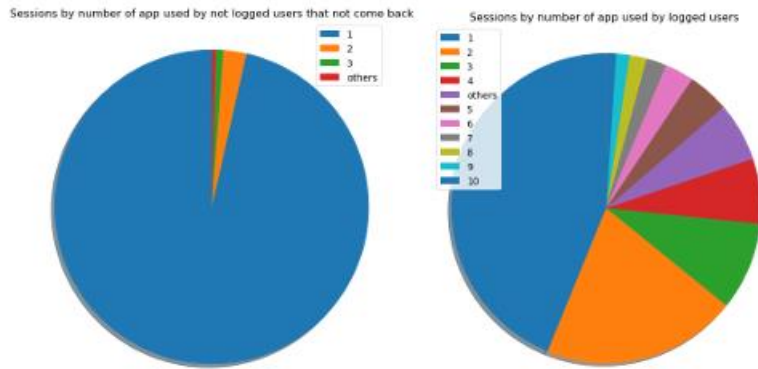
Another interesting aspect to bear in mind is the number of applications used during a session and their duration because the sessions are more informative when using multiple applications as it can be inferred what the user is doing or trying to do.



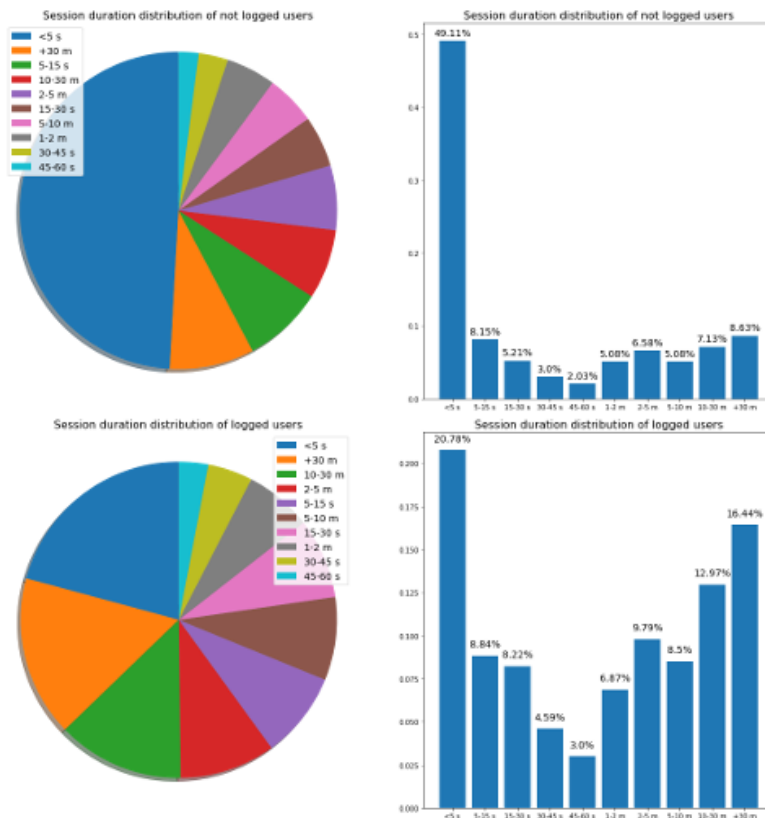
An important aspect to bear in mind when we talking about the number of applications used during a session by a logged user, is that exists an application only to perform the login. For this reason, it is interesting (and important) to evaluate the number of applications of logged users taking into account all the applications except this one (Wipo Account). As we can see below, with this detail, these users remain in general using more applications than the rest.



We can see that expert users tend to use more applications. If we compare the non-logged in users who do not return with logged in users, the percentage of sessions that only use one application drops from 96% to 44%.



In order to analyse the duration of the sessions, it was necessary to calculate the time spent on each application based on the server time of their events. As we can see below, there is large number of sessions with very short or very long durations.



3. Data Cleaning

For each application, we have the events that a user has performed in each of them, such as clicking or loading a new page. Knowing this, we can assume that a common

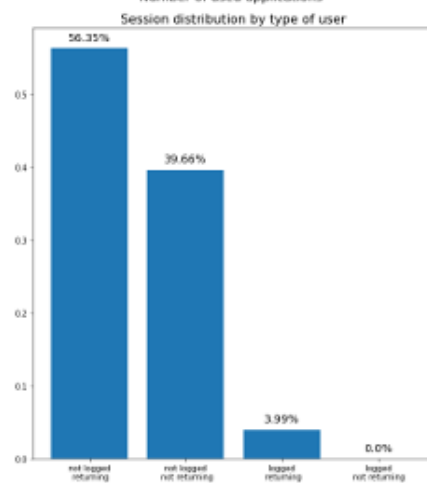
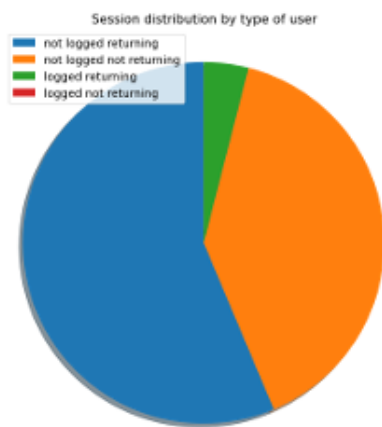
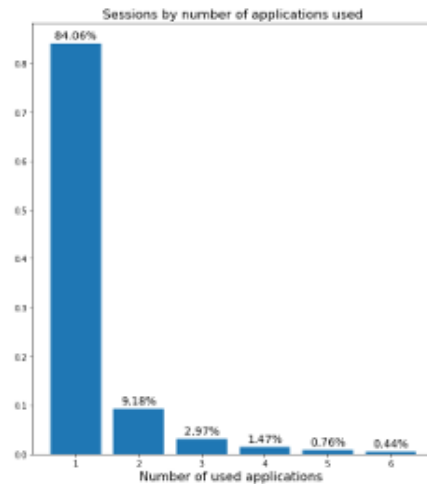
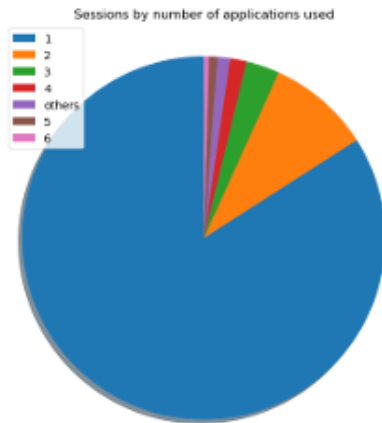
user will perform events with short durations (few seconds or minutes) and we should expect to have sessions with a limited duration (less than a day). By definition, a session ends after 30 minutes of inactivity so an event should not be able to reach that duration.

Based on this, the first rule created to remove undesired sessions is delete sessions with events lasting more than 25 minutes.

1. Ignore sessions with events lasting more than 25 minutes. With this rule, the sessions that contain an event on any application with at least 25 minutes of duration are marked as outliers and ignored. This rule affects to 17.740/488.032 sessions but even with this constraint, there are still many sessions with several days duration.
2. Ignore sessions with at least 6 consecutive events of 10 minutes or more. With this rule, the sessions that contain six or more consecutive events with a duration of 600 seconds are marked as outliers and ignored. This rule affects to 648/488.032 sessions but even with this constraint, there are still many sessions with several days duration.
3. Ignore sessions with only one event with a duration less than a second. With this rule, the sessions that contain only one event with duration less than one second are marked as outliers and removed. This rule affects to 53.738/488.032 sessions but even with this constraint, there are still many sessions with several days duration.

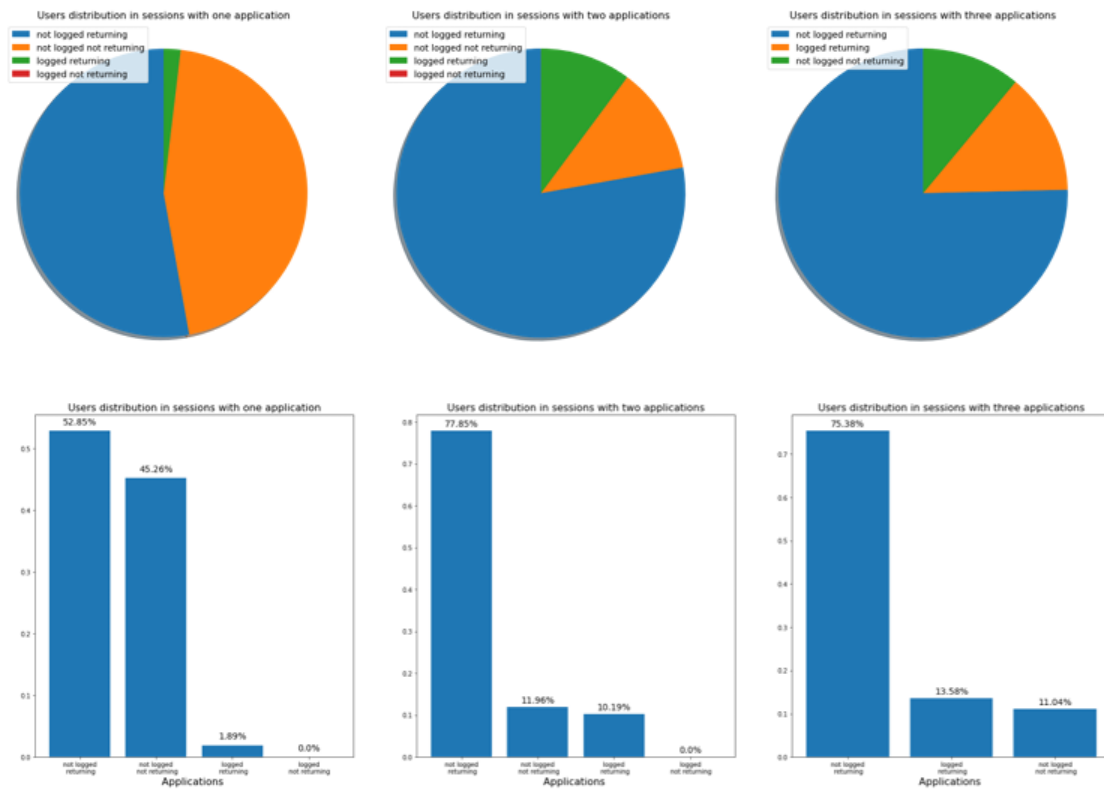
4. Sessions with one application analysis

After remove sessions with non-human behaviour, we can try to select what kind of sessions are useful to try to understand which sessions are more important to understand how familiar users use applications. The distribution of the sessions based on their number of applications and their type of user remains quite similar to the original data. We can see that most of sessions have a single application and the 56.35% of users are not experimented users.



We are interested in having the largest number of sessions that use different applications, but it is also important that these sessions are from users with some experience.

If we take a look to the sessions depending of their number of experimented users, we can see that in the case of sessions with a single app, the 45% of the users are not logged users that not come back. In sessions with more than one application, these users represent 10% or less.



Due to the limited variety of applications, the large number of inexperienced users and the difficulty of knowing what the user is trying to do, we recommend filtering sessions with a single application.

5. Data analysis comparison

In this comparison will be shown the differences between the original dataset, the dataset after apply the filtering rules and the dataset after apply the filtering rules and remove the sessions with a single application.

	Logged users that come back	Logged users that not come back	Not logged users that come back	Not logged users that not come back
Original dataset	19.515 (8.141 unique)	7	276.345 (155.589 unique)	192.165
Filtered dataset	16.211 (7.599 unique)	6	228.832 (137.053 unique)	161.055
Filtered dataset without single application sessions	9.758 (6.234 unique)	3	48.403 (39.468 unique)	6.552

Table 1: Users distribution

	Sessions with 1 application	Sessions with 2 applications	Sessions with 3 applications	Sessions with more than 3 applications
Original dataset	416.621	39.065	13.530	18.816
Filtered dataset	341.388	37.264	12.068	15.384
Filtered dataset without single application sessions	0	37.264	12.068	15.384

Table 2: Sessions distribution by number of applications

